Extended reading — Week 10: SAC, TD3, CrossQ

October 6, 2026

1. Lillicrap, Hunt, Pritzel, Heess, Erez, Tassa, Silver, Wierstra (2015) — Continuous Control with Deep Reinforcement Learning (DDPG)

ICLR 2016. arXiv:1509.02971.

Summary. The first deep RL paper for continuous control. Combines DPG (Silver 2014) with DQN's target networks and replay buffer. The result is a deterministic actor and a Q-critic, both deep nets, trained off-policy.

Why it matters for this week. DDPG is the algorithmic parent of TD3 and (with stochastic policy) of SAC. The replay buffer + polyak-averaged targets + off-policy actor update are all introduced here. Reading this paper makes the structure of TD3 (twin critics, delays) feel like obvious fixes rather than arbitrary tricks.

Annotated key passage.

> *"The fundamental challenge is that the use of large, non-linear function approximators for value learning typically leads to overestimation of the value function."*

This is the observation that motivates everything in TD3. Once you see overestimation as the central problem, every subsequent design (twin critics, target smoothing, conservative-Q in offline RL) makes sense.

2. Fujimoto, van Hoof, Meger (2018) — Addressing Function Approximation Error in Actor-Critic Methods (TD3)

ICML 2018. arXiv:1802.09477.

Summary. TD3. Identifies three pathologies of DDPG (overestimation bias, function-approximation error accumulation, deterministic-policy target sharpness) and provides three fixes (twin critics with min target, delayed actor updates, target policy smoothing). The empirical gain over DDPG is enormous: $\sim 2\times$ final return on every MuJoCo benchmark.

Why it matters for this week. TD3 is the cleanest exposition of the bias-aware actor-critic recipe. Even modern off-policy methods (REDQ, CrossQ, BRO) keep TD3's twin-critic min target as their default.

Annotated key passage.

> *"Taking the minimum of the two Q-functions provides a low-variance estimator of the true Q-function, but with an additional pessimism bias."*

The min trick is a pessimism prior: "trust the more conservative of the two Q estimates". This pessimism is also the principle behind CQL in offline RL (week 14). The trade-off (slightly lower true returns, but no over-estimation cascade) is one of the cleanest in modern RL.

3. Haarnoja, Zhou, Abbeel, Levine (2018a) — Soft Actor-Critic + Haarnoja et al. (2018b) — Automatic Temperature

ICML 2018 / arXiv:1801.01290 + arXiv:1812.05905.

Summary. Two papers, one algorithm. 2018a derives SAC from the maximum-entropy MDP framework, with hand-tuned temperature α. 2018b drops the hyperparameter by introducing the dual variable trick: maintain $\log \alpha$ and update it by gradient descent on a constraint that the policy entropy stays above a target.

Why it matters for this week. The algorithm we implement. Read 2018b end to end; the dual derivation in Appendix C is the cleanest explanation of automatic temperature in the literature.

Annotated key passage.

> *"We propose to optimize this constraint problem via dual gradient descent... the dual update on α becomes $\alpha \leftarrow \alpha - \eta_\alpha \nabla_\alpha \alpha(-\log \pi(a|s) - \mathcal{H}_0)$, leading to a smooth, automatic schedule of the temperature."*

The dual interpretation is the key. Once you accept that "α is a Lagrange multiplier for an entropy constraint", the update rule writes itself — and the resulting algorithm requires no hyperparameter tuning for α.

4. Bhatt, Palenicek, Kuhn, Belousov, Peters (2024) — CrossQ: Batch Normalization in Deep Reinforcement Learning for Greater Sample Efficiency

ICLR 2024. arXiv:1902.05605 (extended in 2023/2024).

Summary. CrossQ. Removes the polyak-averaged target networks of SAC/TD3 by carefully placing BatchNorm: BN after every Linear in both the actor and the critic, with the BN running statistics shared across both the current and next-step Q estimates. The result: $\sim$25% wall-clock improvement and one fewer hyperparameter (the polyak coefficient τ no longer exists).

Why it matters for this week. The most-cited 2024 update to the SAC recipe. Two reasons to read it: (i) target networks are conceptually the most awkward part of SAC, and CrossQ shows they are dispensable; (ii) the BN placement details are non-obvious and the paper is a model of how to do careful empirical ablation.

Annotated key passage.

> *"We find that the stability provided by target networks in SAC and TD3 can be replicated by Batch Normalisation alone, provided the BN layers are placed after every Linear in both actor and critic and the running statistics are updated even when computing target Q-values."*

A simple, almost mundane sentence that encodes a major simplification of modern RL. The mechanism (BN-induced effective learning-rate decay over the recent past) is closely related to the variance-reduction story of GAE; both stabilise by exponentially-weighted smoothing of an internal statistic.

5. Lee, Hwang, Ahn (2025) — SimBa: Simplicity Bias for Scaling Up Reinforcement Learning

ICLR 2025. arXiv:2410.09754.

Summary. SimBa. A drop-in residual + LayerNorm + RMSNorm block that scales SAC/TD3 critics to ∼5M parameters without instability. Pairs with periodic resets, distributional value heads (two-hot/HL-Gauss), and high replay ratios (UTD=10). Beats CrossQ on DMC Dog and Humanoid.

Why it matters for this week. SimBa is the current SOTA off-policy continuous-control baseline. The block construction (Eq. 2 of the paper) is exactly what you would build for synthetic ATS or synthetic insurance if you needed to scale beyond 1M params.

Annotated key passage.

> *"A simple residual block with PreNorm (RMSNorm before each MLP, LayerNorm at the residual sum) suffices to enable stable training of critics with up to 5M parameters and a replay ratio of 10."*

The recipe is no more exotic than that of a small transformer block, applied to actor-critic. It is striking how late it took the RL community to absorb the depth + normalisation lessons that NLP had internalised by 2018.

Sourced from

`research/deeprl_core_sota.md` §3.1–3.7 (SAC, TD3, CrossQ, SimBa, REDQ, DroQ).